

Menu-Driven C Program: Digit Checks

Algorithm, Test Cases, C Program and Expected Results

Aim

To write, compile, execute, and test a menu-driven C program using switch-case that accepts a positive integer and checks either whether all its digits are even or whether all its digits are prime.

Problem Statement

Input a positive integer and display a menu with two operations:

- Check whether all digits of the number are even.
- Check whether all digits of the number are prime. Prime digits are 2, 3, 5, and 7.

Inputs, Data Types and Variables

Variable	Data Type	Purpose	Input/Derived
number	int	Stores the positive integer entered by the user	Input
choice	int	Stores menu choice: 1 or 2	Input
temp	int	Temporary copy of number used for digit processing	Derived
digit	int	Stores the current last digit	Derived
allEven	int	Flag: 1 if all digits are even, otherwise 0	Derived
allPrime	int	Flag: 1 if all digits are prime, otherwise 0	Derived

Algorithm

1. Start.
2. Declare integer variables: number, choice, temp, digit, allEven and allPrime.
3. Display the menu: 1. Check whether all digits are even; 2. Check whether all digits are prime.
4. Input the user's choice.
5. Input a positive integer number.
6. If the number is not positive, display an error message and stop.
7. Use switch(choice) to select the required operation.

8. For case 1, set allEven = 1 and temp = number. Repeat while temp > 0: obtain digit = temp % 10. If digit % 2 != 0, set allEven = 0 and stop checking. Otherwise remove the last digit using temp = temp / 10. After the loop, display whether all digits are even.
9. For case 2, set allPrime = 1 and temp = number. Repeat while temp > 0: obtain digit = temp % 10. If digit is not 2, 3, 5, or 7, set allPrime = 0 and stop checking. Otherwise remove the last digit using temp = temp / 10. After the loop, display whether all digits are prime.
10. For any other choice, display "Invalid choice".
11. Stop.

Test Case Table

Test Case	Choice	Number	Expected Output	Actual Output	Result
TC1	1	12345	All digits are not even	All digits are not even	PASS
TC2	1	24680	All digits are even	All digits are even	PASS
TC3	2	2357	All digits are prime	All digits are prime	PASS
TC4	2	2379	All digits are not prime	All digits are not prime	PASS

C Program

```
#include <stdio.h>

int main() {
    int number, choice, temp, digit;
    int allEven, allPrime;

    printf("MENU\n");
    printf("1. Check whether all digits are even\n");
    printf("2. Check whether all digits are prime\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%d", &number);

    if (number <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    switch (choice) {
        case 1:
            allEven = 1;
            temp = number;

            while (temp > 0) {
```

```

        digit = temp % 10;
        if (digit % 2 != 0) {
            allEven = 0;
            break;
        }
        temp = temp / 10;
    }

    if (allEven)
        printf("All digits are even\n");
    else
        printf("All digits are not even\n");
    break;

case 2:
    allPrime = 1;
    temp = number;

    while (temp > 0) {
        digit = temp % 10;
        if (digit != 2 && digit != 3 &&
            digit != 5 && digit != 7) {
            allPrime = 0;
            break;
        }
        temp = temp / 10;
    }

    if (allPrime)
        printf("All digits are prime\n");
    else
        printf("All digits are not prime\n");
    break;

default:
    printf("Invalid choice\n");
}

return 0;
}

```

Sample Expected Outputs

Choice 1, Number 12345: All digits are not even

Choice 1, Number 24680: All digits are even

Choice 2, Number 2357: All digits are prime

Choice 2, Number 2379: All digits are not prime

Result

The menu-driven C program using switch-case was prepared to check whether all digits of a positive integer are even or prime. The program can be compiled and tested using the prepared test cases, and the actual outputs can be recorded in the test case table.

PROGRAM EXECUTION AND TESTING

Compilation of the C program

```
(kali@kali)-[~/Desktop]
$ gcc hello.c -o hello

(kali@kali)-[~/Desktop]
$
```

Figure 1: Compilation of the C program

Test Case 1: Choice 1, Number 12345 - All digits are not even (PASS)

```
(kali@kali)-[~/Desktop]
$ ./hello
MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
Enter a positive integer: 12345
All digits are not even
```

Figure 2: Test Case 1: Choice 1, Number 12345 - All digits are not even (PASS)

Test Case 2: Choice 1, Number 24680 - All digits are even (PASS)

```
(kali@kali)-[~/Desktop]
$ ./hello
MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
Enter a positive integer: 24680
All digits are even
```

Figure 3: Test Case 2: Choice 1, Number 24680 - All digits are even (PASS)

Test Case 3: Choice 2, Number 2357 - All digits are prime (PASS)

```
(kali㉿kali)-[~/Desktop]
$ ./hello
MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
Enter a positive integer: 2357
All digits are prime
```

Figure 4: Test Case 3: Choice 2, Number 2357 - All digits are prime (PASS)

Test Case 4: Choice 2, Number 2379 - All digits are not prime (PASS)

```
(kali㉿kali)-[~/Desktop]
$ ./hello
MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
Enter a positive integer: 2379
All digits are not prime
```

Figure 5: Test Case 4: Choice 2, Number 2379 - All digits are not prime (PASS)

Final Result: The program was compiled and executed successfully. All four prepared test cases produced the expected outputs; therefore, all test cases PASS.